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Telehomeostasis

Nicotine, Parasitic Manipulation, and the
Persistence Continuum

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Abstract

We argue that the tobacco plant's (*Nicotiana tabacum*) production of nicotine constitutes a *telehomeostatic strategy* within the Kolmogorov Theory (KT) framework: the plant maintains its persistence—its reproduction, range expansion, and political defense—by remotely modifying primate neural circuitry through a single alkaloid molecule. Using the *algorithmic psychodynamics* perspective developed for major depressive disorder (Ruffini, Castaldo et al. 2024), we show that nicotine achieves a *triple lock* on the host agent's decision architecture by simultaneously compromising all three functional modules. The **Objective Function lock** operates through the mesolimbic pathway (VTA→NAc) and the Positive Affect Network: $\alpha 4\beta 2$ nAChR binding triggers dopamine release, D2/D3 receptor downregulation installs a reward-deficiency ratchet, and the agent's valence landscape is reshaped so that only nicotine restores baseline hedonic tone. The **Modeling Engine lock** targets the Default Mode Network (PCC, mPFC, precuneus)—the same regions identified as ME substrates in the MDD framework—rewriting the agent's self-model so that “I am a smoker” becomes a stable identity attractor defended by the comparator. The **Planning Engine lock** follows: optimizing over a hijacked OF and a rewritten self-model, the Cognitive Control Network converges on “smoke” as both reward-maximizing and identity-consistent. We place this triple-lock mechanism on a continuum with canonical parasitic manipulation systems (*Toxoplasma gondii*, *Ophiocordyceps unilateralis*, *Ampulex compressa*), arguing that nicotine is not merely analogous to a parasitic manipulation factor—it *is* one, distinguished only by operating at a distance from the producing organism. We formalize this as *telehomeostasis*: the maintenance of an organism's persistence through a chemical proxy that executes behavioral programs on foreign neural hardware. The paper integrates the self-medication hypothesis (nicotine as anthelmintic in ancestral environments) to show that the same telehomeostatic relationship can flip between mutualism and parasitism depending on ecological context—a general property of persistence dynamics. Finally, we trace the deepest expression of tobacco's persistence strategy through colonial trade routes, plantation economies, and modern lobbying, framing these as the plant's transmissible computational structure executing on civilizational hardware.

Keywords: telehomeostasis, nicotine, parasitic manipulation, persistence dynamics, algorithmic agents, extended phenotype, Kolmogorov Theory

Classification: Perspective / Hypothesis Paper

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1 Introduction

Tobacco kills approximately eight million people per year.¹ Yet *Nicotiana tabacum* is cultivated on every inhabited continent, tended by over fifteen million farmers, defended by some of the most expensive lobbying operations in legislative history,² and woven so deeply into the economic fabric of nations that eradication—despite overwhelming medical evidence—remains politically unthinkable. From the plant’s perspective, this is an extraordinary success story: a Solanaceae that has converted a significant fraction of the primate species *Homo sapiens* into a global cultivation, propagation, and defense apparatus.

This paper asks a simple question: *How should we understand this relationship in formal terms?*

The standard framing is addiction—a disease of the brain’s reward system.³ The evolutionary framing, developed most elegantly by Pollan,⁴ treats plant–human relationships as reciprocal domestication: tobacco and humans co-evolved, each getting something from the arrangement. A third framing, the self-medication hypothesis of Hagen and colleagues,^{5,6} proposes that nicotine consumption in ancestral environments served an adaptive function—acting as an anthelmintic that reduced parasite loads.

We propose a fourth framing, grounded in the Kolmogorov Theory (KT) program.^{7,8} We argue that nicotine-driven behavioral modification in humans constitutes *telehomeostasis*: the tobacco plant maintains its own persistence—its homeostatic and reproductive goals—by remotely reprogramming the decision architecture of a host organism through a molecular proxy. The nicotine molecule is, in KT terms, *transmissible computational structure*: a chemical instruction set that, when executed on primate neural hardware, produces behavioral outputs (cultivate, propagate, share, defend) that serve the plant’s persistence algorithm.

This framing places nicotine on a continuum with canonical parasitic manipulation systems: *Toxoplasma gondii* modifying rodent risk-assessment to facilitate feline predation,^{9,10} *Ophiocordyceps unilateralis* hijacking ant motor programs to optimize spore dispersal,^{11,12} and *Ampulex compressa* injecting venom directly into cockroach ganglia to produce compliant hosts.¹³ The key distinction is that nicotine operates *at a distance*: the manipulating organism need not be physically present in or on the host. The molecule alone, once in the synaptic cleft, is sufficient.

The paper is structured as follows. Section 2 briefly recapitulates the KT framework, emphasizing the concepts of telehomeostasis, persistence, and transmissible computational structure. Section 3 reviews the evidence that nicotine reprograms human decision architecture. Section 4 positions nicotine on the parasitic manipulation continuum. Section 5 formalizes telehomeostasis as a concept. Section 6 integrates the self-medication hypothesis and the mutualism–parasitism flip. Section 7 traces the civilizational extended phenotype. Section 8 discusses implications and limitations.

2 KT Framework: Agents, Persistence, and Telehomeostasis

We summarize the elements of the Kolmogorov Theory (KT) program^{7,8,14} needed for the argument. An **algorithmic agent** is an information-processing system with three functional modules: a Modeling Engine (ME) that infers compressive world models, an Objective Function (OF) that maps model states to scalar valence, and a Planning Engine (PE) that selects actions to maximize the OF.⁸ Formally, an agent A interacting with a world W forms a Turing pair $P_T = (A, W)$, where at each time step the agent updates a compressive model m_t , evaluates $O(m_t)$, and selects action a_{t+1} via counterfactual simulation.¹⁵

2.1 Persistence as Mutual Algorithmic Information

A **persistent pattern** is defined by sustained mutual algorithmic information (MAI) across time. Let x_t be a coarse-grained system description at time t . The instantaneous persistence at lag Δ is:

$$P_\Delta(t) := I(x_t : x_{t+\Delta}) \quad (1)$$

where $I(\cdot : \cdot)$ denotes mutual algorithmic information.¹ A pattern persists when normalized persistence $\tilde{P}_\Delta(t) := I(x_t : x_{t+\Delta}) / \min\{K(x_t), K(x_{t+\Delta})\}$ remains above a threshold θ_Δ over a sustained interval. **Algorithmic Death** is the irreversible collapse of $\tilde{P}_\Delta$ in the algorithmic soup below viability thresholds.

2.2 Telehomeostasis

The OF of biological agents reflects **(tele)homeostatic** goals.⁸ *Homeostasis* is persistence of self—the maintenance of internal variables within a viability set. *Telehomeostasis* extends this: it is persistence of kind, kin, or pattern at a distance. Life, in KT terms, is identified with agenthood—persistent patterns that actively interact with their environment through modeling, planning, and action—as distinguished from passive persistent patterns (a diamond, a crystal) by bidirectional engagement with the world.

2.3 Transmissible Computational Structure

The concept of **transmissible computational structure** bridges biology, AI, and culture.¹⁶ The heritable program H_g at generation g contains the transmissible generative and control structure passed across generations via a transmission operator T . What matters is whether lifetime-acquired modifications can be compiled back into H via a write-back map C . The **write-back vector** $\lambda = (\lambda_B, \lambda_F, \lambda_P)$ parameterizes the bandwidth, fidelity, and persistence of this channel, placing Darwinian ($\lambda \approx 0$) and Lamarckian (large λ) evolution on a continuum.

3 Evidence for Behavioral Reprogramming by Nicotine

We review the mechanisms by which nicotine modifies human decision architecture across multiple levels: molecular, neural-circuit, self-model, social, and genetic. The KT framework provides a natural language for this analysis: the *algorithmic psychodynamics* perspective developed in the context of depression¹⁴ identifies the ME (modeling engine), OF (objective function), and PE (planning engine) as the functional modules through which psychopathology—and, we argue, parasitic manipulation—operates. We show that nicotine’s remarkable effectiveness as a manipulation factor derives from its ability to simultaneously compromise all three modules.

3.1 Objective Function Hijacking: The Reward-Deficiency Ratchet

The primary target is the OF. Nicotine binds to $\alpha 4\beta 2$ nicotinic acetylcholine receptors (nAChRs) on dopaminergic neurons in the ventral tegmental area (VTA), triggering phasic dopamine release in the nucleus accumbens (NAc).^{3,17} This dopamine signal is the brain’s canonical reward-prediction error: it signals that an outcome was better than expected. Nicotine co-opts this signal, tagging the act of smoking as unexpectedly rewarding regardless of its actual consequences for the organism’s persistence.

Chronic exposure produces a compensatory downregulation of striatal D2/D3 dopamine receptors,¹⁸ creating what we term the *reward-deficiency ratchet*. As receptor density falls, the baseline hedonic state shifts downward. Normal rewards—food, social contact, achievement—generate

¹ $I(x : y) = K(x) + K(y) - K(x, y)$, where $K(\cdot)$ is prefix Kolmogorov complexity.

diminished dopamine signals. Only nicotine, which bypasses the receptor downregulation by acting upstream at the VTA, restores baseline hedonic tone. The agent’s OF has been *recalibrated*: the valence landscape now has a deep attractor at “smoking” and flattened gradients elsewhere.

In the algorithmic psychodynamics framework of Ruffini, Castaldo et al.,¹⁴ the neural substrates of the OF include the hypothalamus (HYP), thalamus (TH), paraventricular thalamus (PVT), basal ganglia (BG), ventral striatum (VS)/NAc, amygdala (AMY), subgenual anterior cingulate (sgACC), and orbitofrontal cortex (OFC). Nicotine’s primary target—the VTA→NAc dopaminergic projection—is the *mesolimbic pathway* (MLP), which P10 identifies as a core OF circuit mediating reward prediction error and motivational salience. At the functional-network level, nicotine hijacks the *Positive Affect Network* (PAN: VS, OFC, vmPFC, ACC), whose hypoactivity in depression produces anhedonia; nicotine artificially inflates PAN signaling, creating a pharmacologically maintained hedonic floor. The cortico-thalamo-striato-cortical (CTSC) loop, which P10 maps to both OF and PE, is also implicated: chronic nicotine reshapes habit circuitry in the dorsal striatum, converting voluntary smoking into automatized behavior that persists even when the agent’s explicit PE evaluates it negatively.

The OF modification in smoking is structurally analogous to the persistent low-valence state that characterizes depression, but with a crucial difference: in depression the OF dysfunction is endogenous (arising from the agent’s own circuit pathology), while in smoking the OF dysfunction is *exogenously imposed* by the manipulation factor. The agent’s scalar valence function $O(m_t)$ has been reshaped such that states in which nicotine is present score systematically higher than states in which it is absent, independent of the agent’s original (telehomeostatic) objectives.

3.2 Modeling Engine Manipulation: The Self-Model Rewrite

The modification extends beyond the OF to the agent’s ME, and specifically to the self-model—perhaps the most insidious and theoretically interesting aspect of nicotine’s manipulation strategy. Most addictive substances modify the OF (reward hijacking) and consequently the PE (compulsive seeking). Nicotine does something additional: it rewrites the agent’s model of *who it is*.

Longitudinal studies show that smokers progressively incorporate smoking into their identity.¹⁹ “I am a smoker” becomes a stable attractor in the self-model. Tombor et al.²⁰ demonstrated that non-smoker identity following a quit attempt was the single strongest predictor of long-term abstinence in a population survey—stronger than abstinence self-efficacy, duration of previous quit attempts, or nicotine dependence severity. Conversely, smokers who maintained a smoker identity (framing themselves as “a smoker trying to quit” rather than “a non-smoker”) showed dramatically higher relapse rates. The identity *is* the addiction, at a level that pure pharmacological dependence does not capture.

Why is nicotine especially effective at self-model rewriting? We identify four converging mechanisms:

Frequency and salience. Smoking occurs 10–20 times per day—far more frequently than heroin or cocaine use. Each episode is a self-referential event (“I am choosing to smoke”) that is consolidated into autobiographical memory. Nicotine itself, acting on cholinergic systems that modulate memory consolidation,³ may enhance the encoding of these smoking-related episodic memories, strengthening the self-referential trace with each repetition.

Public identity signaling. Unlike most substance use, smoking is a visible, social, identity-signaling behavior. The smoker’s self-model is continuously reinforced by social feedback loops:

others perceive them as a smoker, treat them as a smoker, and this external recognition is incorporated into the ME’s self-model via the comparator.

Gestural salience and somatic display. We hypothesize that the motor ritual of smoking itself contributes to the ME lock through a mechanism we have not seen discussed in the addiction literature: *attentional capture via gestural display*. Smoking is a repetitive hand-to-mouth movement that draws the observer’s gaze to two body regions of high social and sexual salience—the hands and the mouth. It is difficult not to watch someone smoke: the gesture commands visual attention in a way that, say, swallowing a pill does not. This attentional capture has two telehomeostatic consequences. First, it increases the probability of social transmission: a behavior that is visually salient is more likely to be noticed, encoded, and imitated by observers, facilitating the network-level propagation discussed in Section 3.3. Second, and more subtly, the gestural aesthetics of smoking—the ritualized lighting, the slow exhalation, the hand positions that have been absorbed into cinematic and photographic iconography—associate the act with bodily display and self-presentation, adding a somatic-erotic dimension that the ME integrates into the smoker’s social identity. The smoker is not merely consuming a drug; they are *performing* a socially legible, attention-commanding gesture that becomes part of how they are seen and how they see themselves. This performative dimension may partly explain why smoking has proven so much more culturally persistent than pharmacologically equivalent nicotine delivery systems (patches, gums, lozenges) that lack the gestural component: the ME lock requires the visible ritual, not just the molecule.

Default mode network remodeling. Chronic smoking alters functional connectivity in the default mode network (DMN)—the medial prefrontal cortex (mPFC), posterior cingulate cortex (PCC), angular gyrus (AG), and precuneus (PCu)—which are the neural substrates of self-referential processing.²¹ Crucially, P10’s translational framework identifies exactly these regions as the neural substrates of the Modeling Engine: AG (semantic processing, self-referential thinking), PCC (DMN hub), PCu (self-referential processing, spatial cognition), and the hippocampus (HC; episodic memory, context representation).¹⁴ The PCC↔HC circuit, which P10 flags as a core ME pathway, is the circuit through which smoking-related episodic memories are consolidated and linked to self-referential representations. Smoking cues reliably engage DMN regions during cue-induced craving,²² embedding smoking-related representations into the self-referential processing stream. This creates a bidirectional feedback loop: smoker identity activates DMN smoking representations, which generate craving, which triggers smoking, which reinforces the identity. In P10’s terminology, nicotine produces a *Rumination/Inattention*-like pattern in the ME—but instead of negative rumination (as in depression), the content is self-referential smoking identity that the comparator defends against revision.

Autonomous internalization. Research grounded in self-determination theory²³ shows that successful cessation requires not just behavioral change but autonomous identity change—the agent must come to see “non-smoker” as its true self, not as an externally imposed constraint. This implies that the smoker identity was itself autonomously internalized: the agent does not experience smoking as imposed (as it might experience, say, a compulsion), but as *who it genuinely is*. The manipulation has achieved the deepest possible integration into the ME’s self-model.

In KT terms, this represents a qualitatively different kind of parasitic manipulation from anything seen in the canonical systems. *T. gondii* modifies the host’s OF (fear response). *O. unilateralis* hijacks the PE (motor programs). But nicotine rewrites the ME’s self-model—the agent’s compressed description of its own identity. Cessation attempts are experienced not merely as

withdrawal (OF disturbance) or as inhibiting a compulsion (PE struggle), but as *losing a part of the self* (ME prediction error). This is why smoking is so difficult to quit despite being, pharmacologically, a moderate dependence: the manipulation has been written into the agent's model of its own identity, and the comparator defends that model against revision.

Remark (Algorithmic Psychodynamics of Nicotine Manipulation)

The algorithmic psychodynamics perspective¹⁴ reveals that nicotine achieves a *triple lock* on the host agent's decision architecture, mapping precisely onto the agent-to-circuit architecture established in P10:

- (i) **OF lock** (circuits: MLP/VTA→NAc, PAN, CTSC): reward-deficiency ratchet (D2/D3 downregulation in ventral striatum) ensures that only nicotine restores baseline valence. This targets the same mesolimbic and positive-affect circuitry whose hypoactivity produces anhedonia in depression—but here the dysfunction is exogenously imposed;
- (ii) **ME lock** (circuits: DMN/PCC↔HC, AG, PCu): self-model rewrite (“I am a smoker”) ensures that cessation generates identity-level prediction error. The DMN regions that P10 identifies as ME substrates—PCC, mPFC, AG, PCu, HC—are remodeled by chronic smoking to incorporate smoking identity into the self-referential processing stream;
- (iii) **PE lock** (circuits: CCN/dlPFC, CTSC, prefrontal-amygdala): the PE, optimizing over an OF that scores smoking highly and an ME that predicts “I smoke,” converges on the action “smoke” as both reward-maximizing and identity-consistent. The Cognitive Control Network (dlPFC, posterior parietal cortex), which P10 maps to the PE, shows reduced capacity to override the habituated smoking response—analogueous to the cognitive dyscontrol biotype in depression.

All three modules reinforce the same behavioral output. This triple lock explains why nicotine dependence is so robust despite the molecule's moderate pharmacological potency: the manipulation is distributed across the entire agent architecture, not concentrated in a single module. The cross-link with P10's MDD biotypes is instructive: depression and nicotine addiction can be understood as *dual dysregulations* of the same agent circuitry—endogenous in depression, exogenously imposed in addiction—which explains their high comorbidity and shared circuit-level signatures. Human self-report—unavailable for ant or rodent hosts—can in principle disambiguate which module is primary in a given individual's addiction, opening a path to biotype-specific intervention strategies analogueous to those proposed for MDD in P10.

3.3 Social Contagion: Network-Level Propagation

Christakis and Fowler demonstrated that smoking behavior propagates through social networks with quantifiable transmission probabilities: smoking cessation by a spouse reduces the partner's probability of smoking by 67%; by a friend, 36%; by a sibling, 25%.²⁴ The converse also holds: smoking propagates through the same channels. This is not merely social influence—it is the behavioral output of nicotine's manipulation spreading to new hosts via social learning, precisely as a parasitic lifecycle requires secondary transmission.

From the plant's perspective, the human social network is a transmission medium. The molecule reprograms one host, who then (through social modeling, cigarette sharing, cultural normalization) exposes others to the molecule. The social contagion data²⁴ can be read as an epidemiological curve of a behavioral parasite propagating through a contact network.

3.4 Genetic Co-Evolution: Selection on Host Susceptibility

Twin studies estimate the heritability of smoking initiation at 37–55% and of nicotine dependence at 59–75%.²⁵ Genome-wide association studies have identified the CHRNA5-CHRNA3-CHRNA4 nicotinic receptor gene cluster on chromosome 15 as a major susceptibility locus, with the non-synonymous rs16969968 variant in CHRNA5 conferring substantially elevated risk.

This heritability is significant from the telehomeostatic perspective. It means that nicotine’s behavioral effects are not uniform across the host population—some genotypes are more susceptible than others. If we frame nicotine as a manipulation factor, then the host population is under selection pressure for susceptibility alleles to the extent that the manipulation relationship was (in some environments) mutualistic (see Section 6). The high heritability of susceptibility suggests a long co-evolutionary history in which the plant’s alkaloid and the primate’s receptor have been shaping each other.

4 The Parasitic Manipulation Continuum

Parasitic manipulation of host behavior is a well-established phenomenon in which a parasite alters its host’s actions in ways that increase parasite fitness, often at the host’s expense.^{26,27} The concept of a **manipulation factor**—a specific molecule produced by the parasite that causally mediates the behavioral change—has been formalized by Herbison et al.²⁸ We now position nicotine on this continuum.

4.1 Canonical Systems

Toxoplasma gondii → **Rodents**. The protozoan *T. gondii* infects rodent brains, specifically targeting the amygdala, and converts innate aversion to cat urine into attraction—a behavioral change that increases the probability of predation by the parasite’s definitive host (felids).^{9,10} The manipulation factor is believed to involve altered dopamine signaling and localized inflammation in fear-processing circuits. The parasite is physically present in brain cysts.

Ophiocordyceps unilateralis → **Ants**. The fungus *O. unilateralis* infects carpenter ants and, at a precise stage of its lifecycle, takes control of the ant’s motor program, causing it to climb vegetation and clamp its mandibles onto a leaf vein at an optimal height and orientation for spore dispersal.¹¹ De Bekker et al. showed that the fungus secretes species-specific arrays of neuromodulatory metabolites—including GABA and sphingosine—in response to target host brain tissue.¹² The manipulation is exquisitely targeted: the fungus can infect multiple ant species but only manipulates behavior in its co-evolved host.

Ampulex compressa → **Periplaneta americana**. The jewel wasp injects venom directly into the cockroach’s subesophageal ganglion and brain, producing a compliant “zombie” state in which the cockroach can walk but will not initiate escape behavior.¹³ The venom contains dopamine, GABA, and other neuroactive compounds. The wasp then leads the cockroach to its burrow, where the wasp’s larva consumes the living host.

4.2 *Nicotiana tabacum* → *Homo sapiens*

Table 1 compares these systems across five dimensions: the manipulation factor, the transmission route, the neural target, the behavioral output, and the fitness payoff to the manipulator.

The critical observation is that nicotine satisfies every criterion for a manipulation factor as defined by Herbison et al.²⁸—it is a specific molecule produced by the manipulating organism that

Table 1: Parasitic manipulation continuum. In each system, a manipulating organism produces a factor that alters host behavior to increase manipulator fitness.

Dimension	<i>T. gondii</i>	<i>O. unilateralis</i>	<i>A. compressa</i>	<i>N. tabacum</i>
Factor	Altered DA / inflammation	GABA, sphingosine	Venom cocktail (DA, GABA)	Nicotine ($\alpha 4\beta 2$ agonist)
Route	Ingestion $\rightarrow$ cyst in brain	Spore $\rightarrow$ systemic infection	Direct injection into CNS	Inhalation $\rightarrow$ blood $\rightarrow$ VTA
Target	Amygdala (fear)	Motor circuits	Subesoph. ganglion	VTA–NAc reward circuit
Output	Reduced cat avoidance	“Death grip” climb	Loss of escape drive	Cultivate, share, defend
Payoff	Transmission to feline	Optimal spore dispersal	Larval food source	Cultivation, range expansion
Proximity	In host tissue	In host tissue	Transient contact	<i>At a distance</i>

causally mediates behavioral change in the host, resulting in increased manipulator fitness—with one structural difference: the manipulator need not be physically present in the host. Nicotine is, following Herbison et al.’s terminology, a *remote manipulation factor*.

This is not merely an analogy. The Herbison et al. framework²⁸ calls for identifying “the missing link”—the molecular mechanism connecting parasite presence to host behavioral change. For nicotine, the link is remarkably well-characterized: $\alpha 4\beta 2$ nAChR binding $\rightarrow$ VTA dopamine release $\rightarrow$ NAc reward signal $\rightarrow$ reinforcement of smoking behavior $\rightarrow$ continued cultivation and propagation of the plant.^{3,17} The causal chain from plant gene (nicotine biosynthesis) to host behavior (agricultural and social propagation) is more completely mapped than for any of the canonical parasitic systems.

5 Telehomeostasis Formalized

We now formalize the concept of telehomeostasis in KT terms, distinguishing it from Dawkins’ extended phenotype²⁹ and from standard parasite–host models.

Definition 1 (Telehomeostatic Manipulation)

Let A be an algorithmic agent (the *manipulator*) with persistence metric $\tilde{P}_\Delta^A(t)$ and let B be a distinct algorithmic agent (the *host*). A engages in **telehomeostatic manipulation** of B if A produces a transmissible factor ϕ such that:

- (i) ϕ modifies B ’s objective function O_B , modeling engine ME_B , or planning engine PE_B (or any combination);
- (ii) the modified behavior of B increases A ’s persistence: $\tilde{P}_\Delta^A(t \mid \phi) > \tilde{P}_\Delta^A(t \mid \neg\phi)$;
- (iii) ϕ operates without requiring A ’s physical co-location with B (the “tele” condition).

Condition (iii) distinguishes telehomeostatic manipulation from standard parasitic manipulation, where the manipulator is physically present in or on the host. In the nicotine case, the plant produces the molecule; the molecule travels through the supply chain (cultivation $\rightarrow$ curing $\rightarrow$ manufacture $\rightarrow$ sale $\rightarrow$ inhalation); and the behavioral effects execute on neural hardware that may be thousands of kilometers from any living tobacco plant.

Definition 2 (Transmissible Computational Structure as Manipulation Factor)

A manipulation factor ϕ is **transmissible computational structure** if, when presented to the execution substrate of host B , it deterministically or stochastically modifies B 's computational state in a way that can be described as executing a program π_ϕ on B 's hardware, where π_ϕ was “authored” by A 's evolutionary history (i.e., π_ϕ is part of A 's heritable program H_g^A).

Nicotine fits this definition precisely. The nicotine biosynthesis pathway is encoded in the tobacco plant's genome (heritable program H_g^A). The molecule, when it reaches the primate VTA, executes a program: bind $\alpha 4\beta 2$ receptors $\rightarrow$ open ion channels $\rightarrow$ depolarize DA neurons $\rightarrow$ release dopamine in NAc $\rightarrow$ modify reward prediction error $\rightarrow$ reinforce preceding behavior. The “program” π_ϕ is the sequence of biochemical and neural consequences that the molecule's structure dictates. The “author” is natural selection acting on the tobacco plant's alkaloid biosynthesis genes.

5.1 Distinction from Dawkins' Extended Phenotype

Dawkins' extended phenotype²⁹ is a descriptive framework: a gene's effects extend beyond the organism's body. Our framing adds three elements. First, it is grounded in a formal theory of agents and persistence (KT), not merely in gene-level selection. Second, it emphasizes *active computational reprogramming*—the manipulation factor does not merely alter the environment (as a beaver dam does) but rewrites the host's decision-making software. Third, it identifies the specific modules at which the manipulation operates (the OF, ME, and PE of the host agent), providing a mechanistic account of *how* the extended phenotype is realized—and, in the case of nicotine, revealing the remarkable triple-lock strategy by which a single molecule compromises all three modules simultaneously.

6 The Mutualism–Parasitism Flip

The self-medication hypothesis of Hagen, Sullivan, and colleagues^{5,30} proposes that human consumption of plant neurotoxins, including nicotine, may have served an adaptive function in ancestral environments: these compounds are potent anthelmintics that reduce intestinal parasite loads. Roulette et al.⁶ provided striking empirical support in a study of Aka foragers in the Congo basin. Among 206 Aka men, salivary cotinine (a nicotine metabolite) was significantly negatively correlated with intestinal helminth burden. Critically, treatment with albendazole (a pharmaceutical anthelmintic) reduced cotinine levels at two-week follow-up, suggesting that the drive to consume nicotine diminished when the parasites it was treating were removed.

This finding transforms the telehomeostatic picture. In helminth-rich environments, the nicotine–human relationship is *mutualistic*: the plant gains cultivation and propagation, and the human gains deworming. The human's OF is modified (nicotine is rewarding), but the modification serves the human's persistence as well as the plant's. In helminth-poor environments—such as modern industrialized societies—the same telehomeostatic mechanism becomes *parasitic*: the OF modification persists (nicotine is still rewarding, D2/D3 downregulation still occurs), but the fitness benefit to the human has vanished. The agent is running a program that was adaptive in a different ecological context but is now purely exploitative.

Remark 1 (Context-Dependent Valence of Telehomeostasis)

The mutualism–parasitism flip is a general property of persistence dynamics in KT terms. Any telehomeostatic relationship—one organism maintaining its persistence by modifying another’s behavior—can shift between mutualism and parasitism as the ecological context changes. What matters is whether the OF modification in the host also serves the host’s persistence. The *mechanism* (nicotine binding, dopamine release, reward recalibration) is identical in both regimes; only the fitness consequences differ.

This has a formal parallel in the write-back framework of WP0058.¹⁶ The nicotine molecule is part of the plant’s heritable program H_g^A . When it modifies the human host’s OF, it is effectively writing information into the host’s plastic runtime state (tier 4 of the four-tier ontology). In helminth-rich environments, this write-back is *cooperative*—both agents’ persistence is enhanced. In helminth-poor environments, the write-back is *exploitative*—only the plant’s persistence benefits. The high heritability of nicotine susceptibility (40–75%)²⁵ suggests that in environments where the mutualistic regime dominated, there was positive selection on host genotypes that facilitated the write-back—a co-evolutionary lock-in that now, in the parasitic regime, manifests as genetic vulnerability to addiction.

7 The Civilizational Extended Phenotype

The deepest expression of tobacco’s telehomeostatic strategy is not in the synaptic cleft but in the structures of human civilization that the molecule has called into existence.

Nicotiana tabacum was domesticated approximately 8,000 years ago in the Andes.³¹ By the time of European contact, it was cultivated and ritually consumed across the Americas. Within a century of Columbus’s arrival, tobacco had become a driver of colonial expansion: the Virginia colony’s survival was predicated on tobacco cultivation; the plantation economy it created became a pillar of the transatlantic slave trade; and the wealth it generated funded the political architecture of the early American republic.³¹

In KT terms, this civilizational trajectory is the plant’s heritable program H_g^A expressing itself through an extraordinary chain of transmissions. The nicotine biosynthesis genes encode the molecule. The molecule modifies human reward circuits. The modified humans build economic systems optimized for tobacco production and distribution. Those economic systems, in turn, create political structures (lobbying organizations, agricultural subsidies, trade agreements) that defend and expand tobacco cultivation.²

This is not metaphor. The causal chain is traceable: plant genome → nicotine biosynthesis → alkaloid in leaf → reward-circuit modification in human brain → cultivation behavior → trade networks → political lobbying → agricultural policy → expanded cultivation. At every link, the information content of the plant’s persistence strategy is being transmitted and amplified. The lobbying firm defending tobacco interests in Washington is, in a precise KT sense, part of the tobacco plant’s persistence apparatus—an extension of its telehomeostatic reach, operating on civilizational hardware.

The modern tobacco industry produces approximately 6 million metric tons of leaf annually, generates over \$750 billion in product sales, and returns nearly \$400 billion in tax revenue to governments.¹ These numbers represent the scale of the plant’s *civilizational phenotype*—the infrastructure that the persistence algorithm, executing on hundreds of millions of human neural substrates, has constructed.

Remark 2 (The Persistence Algorithm Across Substrates)

The progression from alkaloid biosynthesis to civilizational infrastructure illustrates a key KT principle: computational structure is substrate-independent. The “program” that maintains tobacco’s persistence executes on biochemical substrates (the nicotine biosynthesis pathway), neural substrates (the human reward circuit), cognitive substrates (the smoker’s self-model), social substrates (network transmission of smoking behavior), economic substrates (the tobacco industry), and political substrates (lobbying and regulatory capture). The program is the same—maintain and expand *N. tabacum* cultivation—but it runs on hardware spanning molecular to civilizational scales.

8 Discussion**8.1 What This Framing Adds**

The telehomeostatic framing adds three things to the existing literature on nicotine addiction.

First, it provides a *formal mechanism* for the plant’s-eye-view that Pollan⁴ articulated narratively. Where Pollan described reciprocal domestication, we identify the specific computational operation (OF modification via dopamine hijacking) and the specific KT category (telehomeostasis) that makes the relationship work. The framing is not merely “plants use us”—it is “plants maintain their persistence by running programs on our neural hardware, and here is how the program executes.”

Second, it places nicotine on a *continuum with parasitic manipulation* rather than treating addiction as a uniquely human pathology.^{26–28} The mechanisms are conserved: all four systems in Table 1 operate by modifying host neurotransmitter signaling. The distinction between nicotine and Toxoplasma cysts is *proximity*, not *principle*. This reframing connects addiction neuroscience to parasitology in a way that may generate testable predictions about shared molecular mechanisms across manipulation systems.

Third, it integrates the self-medication hypothesis^{5,6} within a framework that explains *why* the same molecule can be mutualistic in one context and parasitic in another, without invoking different mechanisms. The mechanism (OF modification via reward-circuit manipulation) is invariant; the fitness consequences (mutualism vs. parasitism) depend on the ecological context.

8.2 Limitations and Caveats

Several important caveats apply. (i) *N. tabacum* is an algorithmic agent in the KT sense—a minimal one, but an agent nonetheless. Plants sense light gradients (phototropism), gravity (gravitropism), soil chemistry (chemotropism), herbivore damage (jasmonate signaling cascades), and neighboring plants (shade avoidance via phytochrome ratio detection). They allocate resources between growth, defense, and reproduction based on these inputs. They are closer to thermostats than to primates on the agency spectrum surveyed in WP0062, but they satisfy the core criteria: compressive environmental modeling (however implicit), non-trivial objective (persistence, reproduction), and action selection (differential resource allocation, volatile signaling). The telehomeostatic relationship described here is therefore genuinely *inter-agent*: two algorithmic agents of vastly different complexity, where the simpler one has evolved a manipulation factor that reprograms the more complex one’s decision architecture to serve the simpler one’s persistence. (ii) The analogy with parasitic manipulation is strongest at the molecular and neural levels. At the social and civilizational levels, human agency, culture, and institutional dynamics introduce complexities that the simple manipulation-factor model does not capture. (iii) The self-medication

hypothesis, while supported by the Aka forager data,⁶ remains contested and has not been replicated across multiple populations. (iv) The KT formalization (Definitions 1–2) is presented as a conceptual framework, not a quantitative model. Empirical operationalization would require specifying computable proxies for persistence $\tilde{P}_\Delta$ and for the OF modification induced by nicotine.

8.3 Broader Implications

If the telehomeostatic framing is correct, it suggests that other plant alkaloids (caffeine, THC, psilocybin, cocaine) may similarly constitute remote manipulation factors whose mutualism–parasitism balance depends on ecological context. More broadly, any molecule that modifies the OF or PE of a host agent to increase the producer’s persistence qualifies as telehomeostatic computational structure. This opens a research program at the intersection of neuropharmacology, parasitology, and algorithmic information theory.

The framework also speaks to the KT program’s open question Q5 (multi-agent telehomeostasis). The tobacco–human system is a case where a minimal algorithmic agent—one with implicit environmental modeling, resource-allocation action selection, and a persistence-driven objective function—maintains its fitness by co-opting the decision architecture of a vastly more complex algorithmic agent. The relationship is profoundly asymmetric: the human is a high-complexity KT agent with explicit ME, OF, and PE modules, while the plant is a minimal agent (closer to the thermostat end of the agency spectrum) whose “strategy” is encoded in its genome and executed through chemical intermediaries. This asymmetry—a simple agent reprogramming a complex one via a molecular manipulation factor—may be a general feature of cross-kingdom telehomeostatic relationships and of host–parasite dynamics more broadly.

9 Conclusion

We have argued that *Nicotiana tabacum*’s production of nicotine constitutes a telehomeostatic strategy: the plant maintains its persistence by remotely reprogramming the decision architecture of *Homo sapiens* through a single alkaloid molecule that modifies the host’s objective function. This places nicotine on a continuum with canonical parasitic manipulation factors, distinguished by operating at a distance from the producing organism. The same mechanism can be mutualistic (ancestral anthelmintic context) or parasitic (modern industrialized context), depending on the ecological setting—a general property of persistence dynamics in KT terms. The deepest expression of this telehomeostatic strategy is the civilizational infrastructure—trade routes, plantation economies, lobbying organizations, agricultural policy—that the persistence algorithm, executing on human neural hardware, has built.

The nicotine molecule is transmissible computational structure. The human brain is the execution substrate. The behavioral output—cultivate, propagate, share, defend—is the plant’s persistence algorithm running on foreign hardware. This is agent dynamics operating across kingdom boundaries.

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